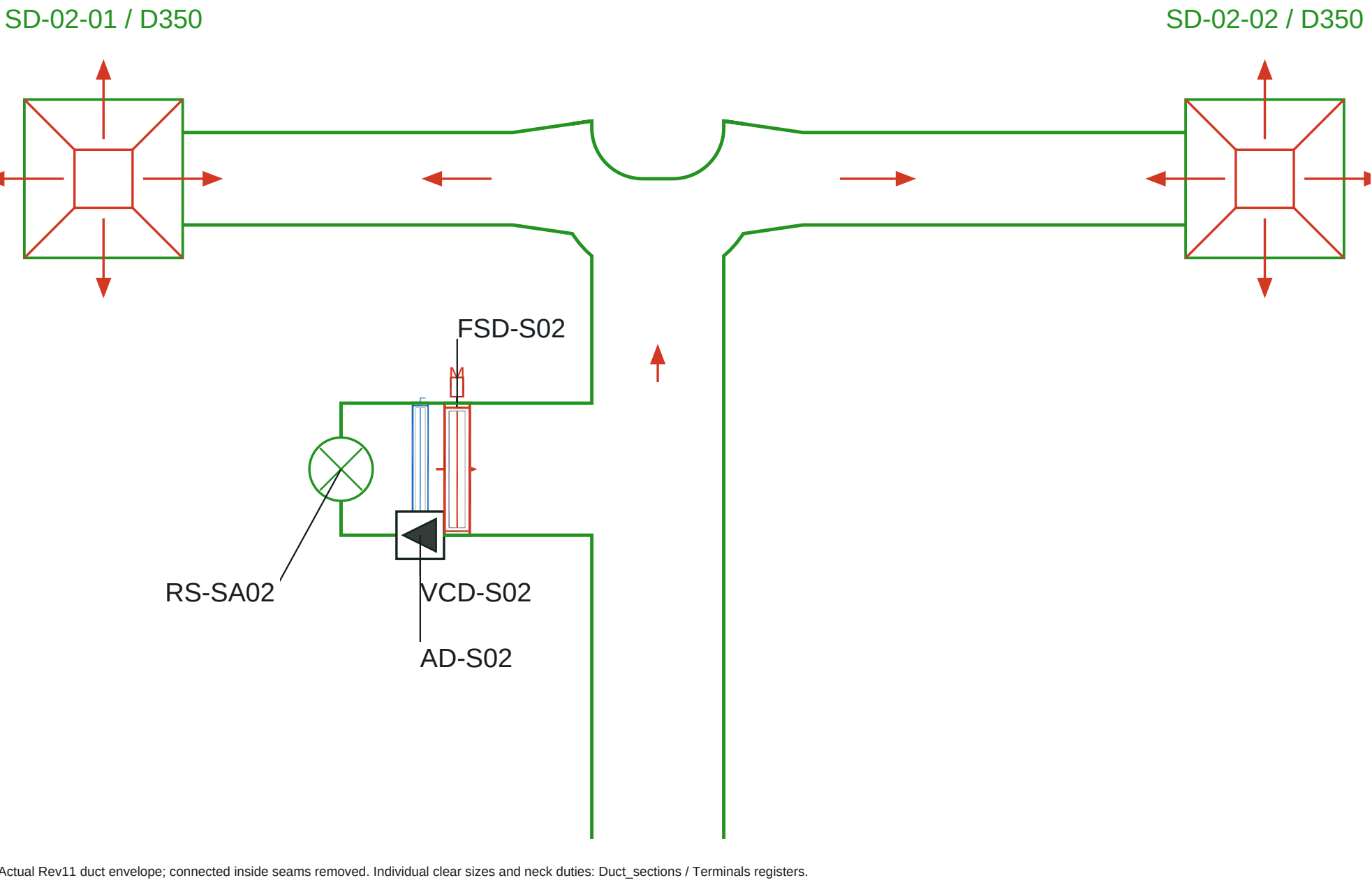


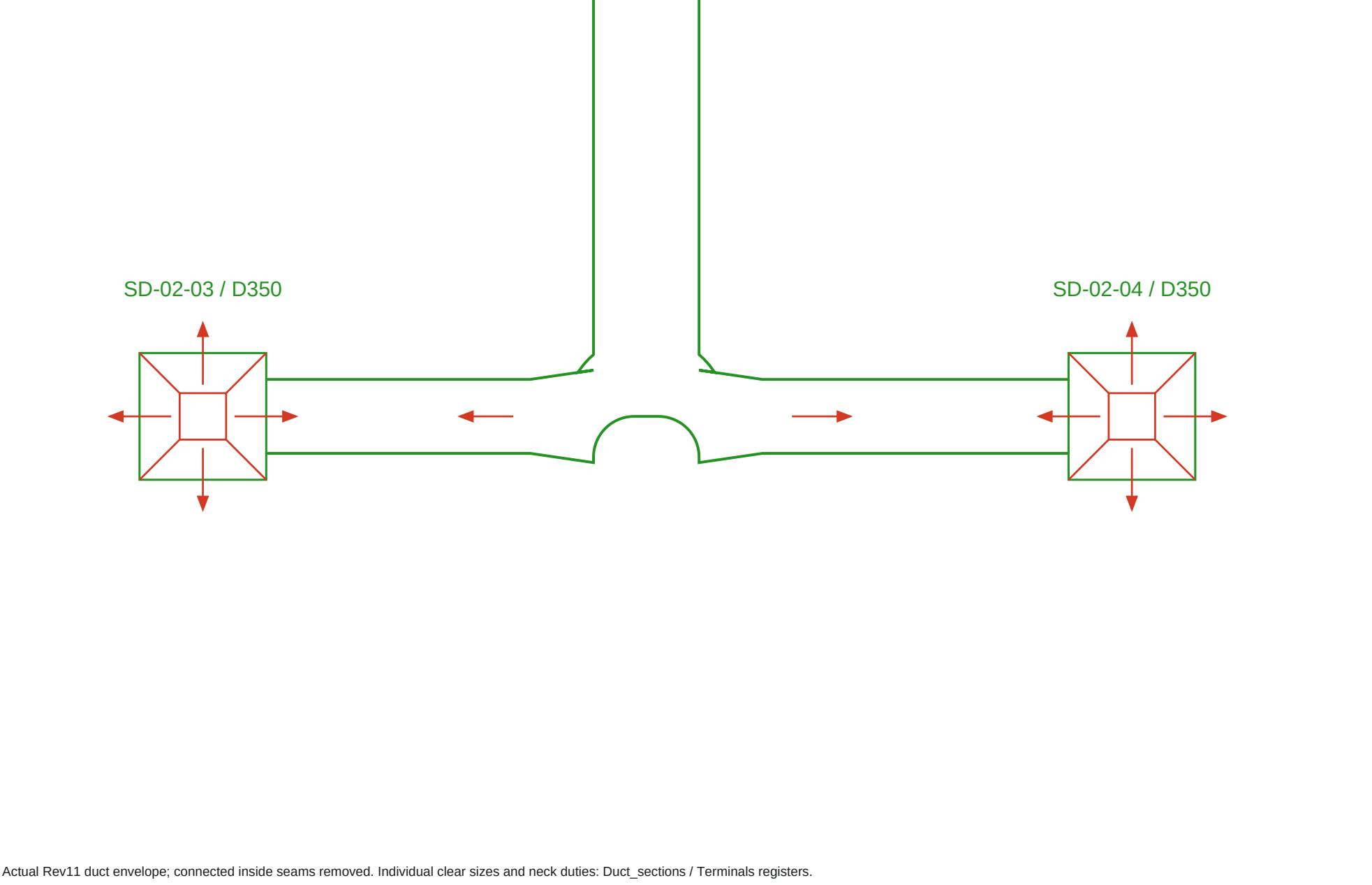
ADMINISTRATION BUILDING - ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

A GF-02 ROOF DROP / BRANCH / HEADER LEGS / 1:20



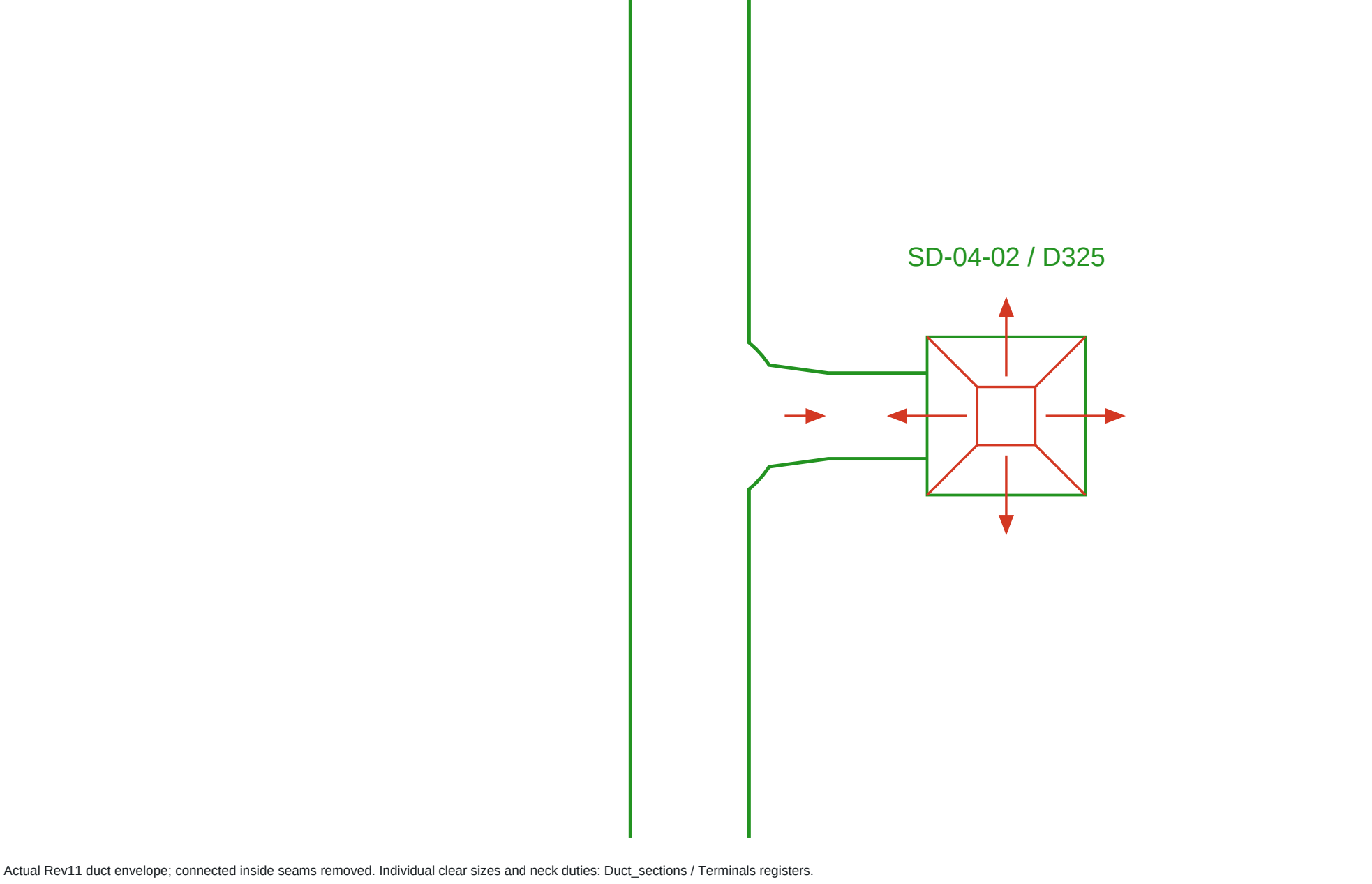
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

B GF-02 END TEE / TWO INDIVIDUAL NECKS / 1:25



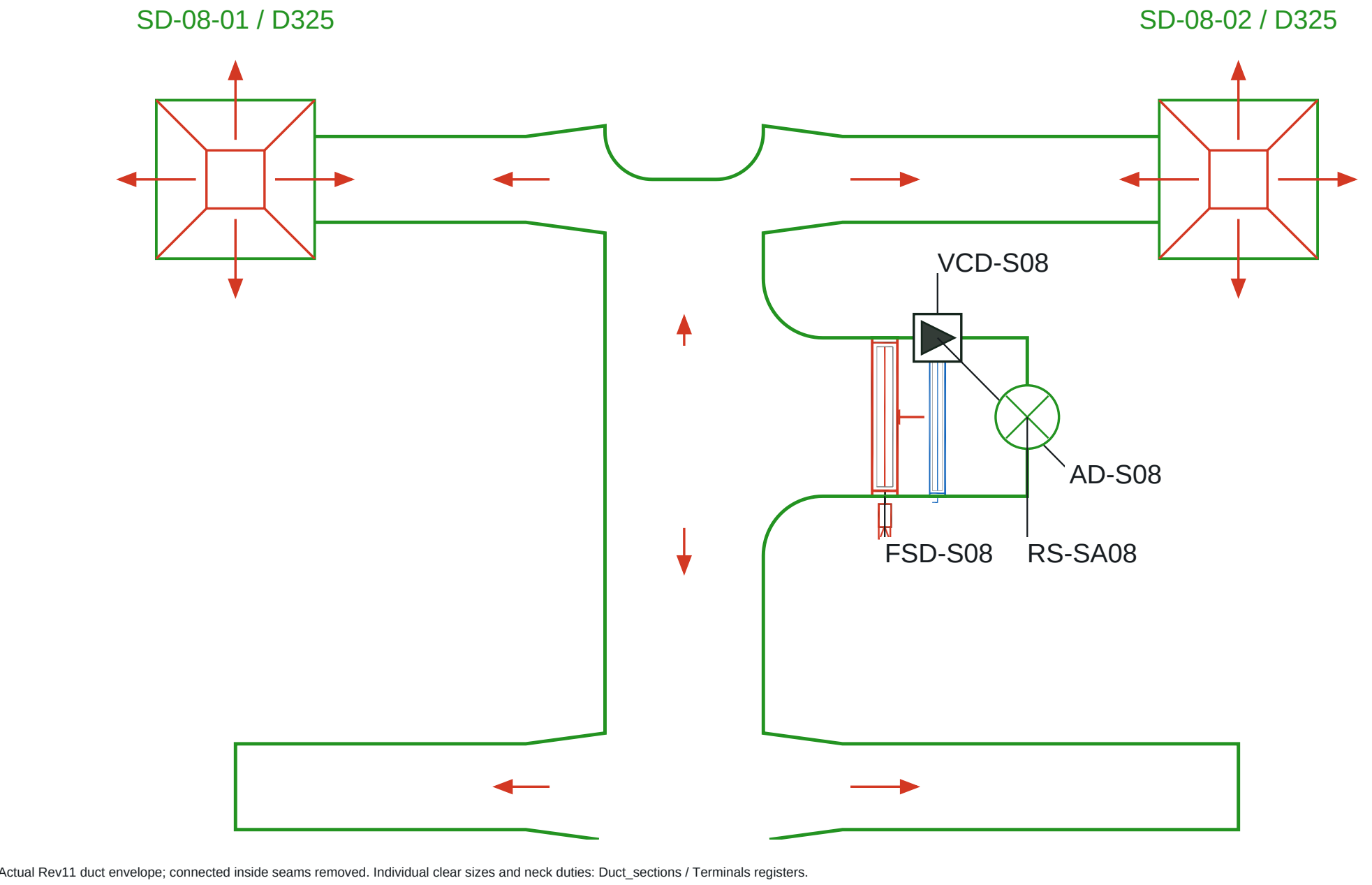
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

C MEETING CONTINUING HEADER / SIDE NECK / 1:20



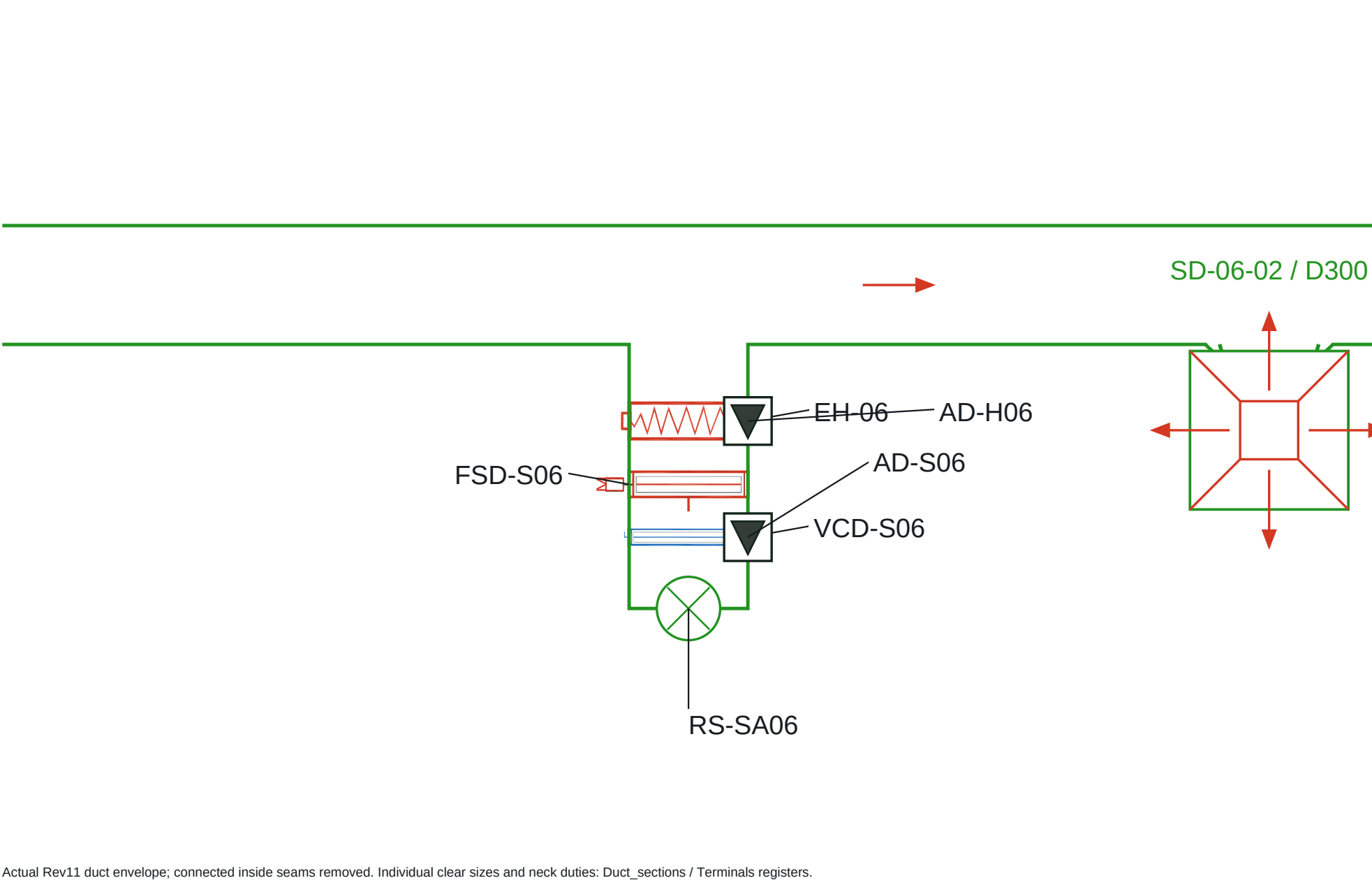
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

D TELECOM FEEDER / UP AND DOWN HEADERS / 1:20



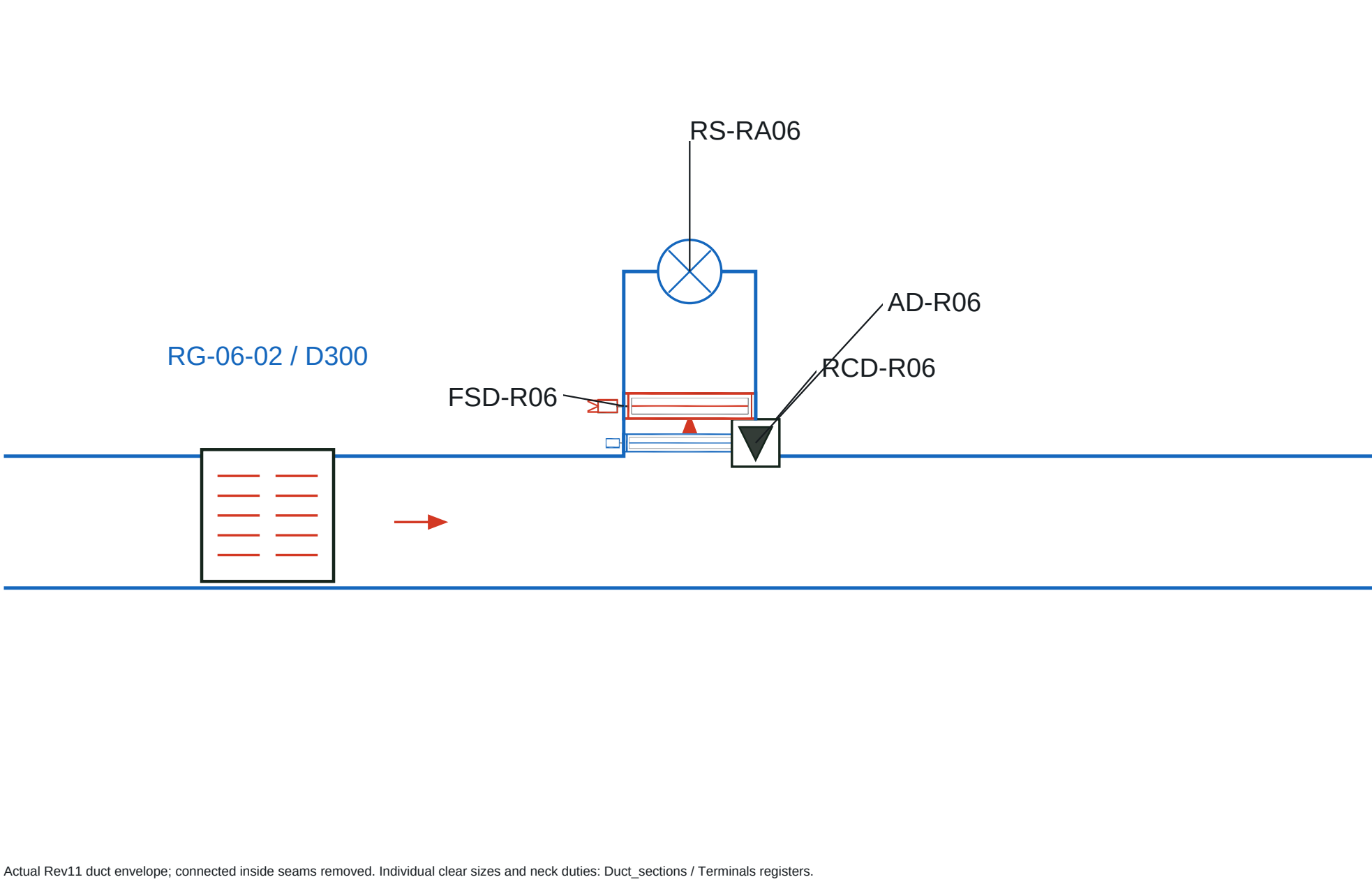
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

E CORRIDOR ROOM DROP BRANCH / HEATER / HEADER / 1:20



Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

F CORRIDOR COLLECTION / RETURN DEVICE STATION / 1:20



Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

These details are generated from the delivered routing geometry, not independent typical symbols. Continuous exteriors, flared necks and concave fitting throats are checked. Short necks use an integral plenum / vertical drop; their selected construction is pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-019

SCALE: AS SHOWN AT A1

REVISION

11

07 OCT 2026

SHEET 19 OF 22

ADMINISTRATION BUILDING - 3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

USER DATUMS AND PROPOSED INDOOR LEVELS

Item	Value	Basis / status
Structural underside	+4.000 m AFFL	User confirmed slab / roof
Minimum ceiling / clear height	+3.300 m AFFL	User requirement
Available total space	700 mm	Between confirmed datums
Proposed bare duct bottom	+3.400 m AFFL	All indoor trees; not installed
Indoor insulation	50 mm each face	Assumed; project specification TBC
Largest bare duct depth	400 mm	Round necks max D350 retained
Lowest insulated bottom	+3.350 m AFFL	50 mm above required ceiling
Highest insulated top	+3.850 m AFFL	150 mm below slab for deepest duct

WHAT THE REVISED ROUTES RESOLVE

Deep common mains move to a proposed roof distribution system. Fifteen independent room supply / return drops replace the indoor common-main feeders.

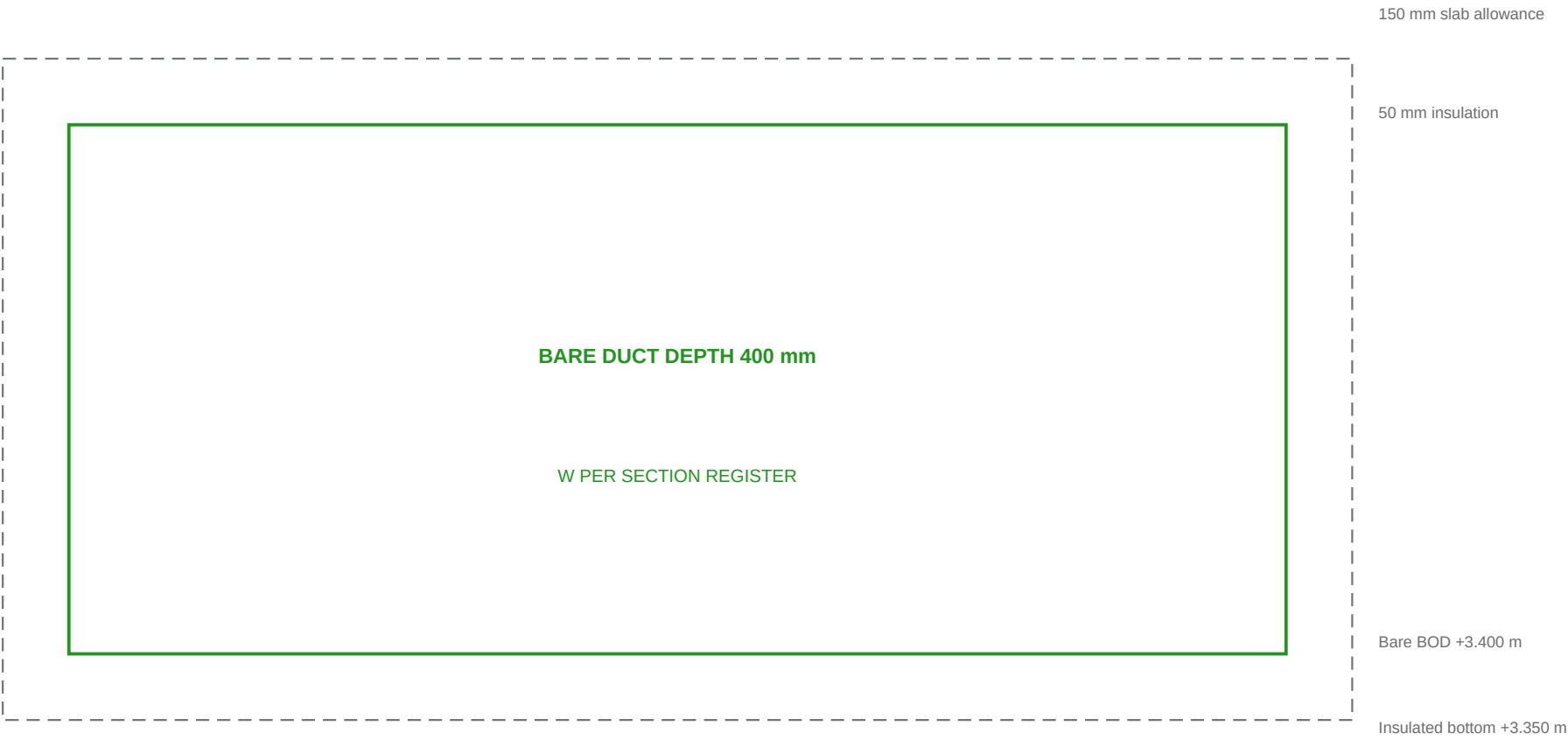
Electrical and telecom returns collect at room perimeters. Every room tree is connected, every collection node conserves airflow, and distinct room trees stay separate.

No indoor bare duct crossings or insulated separate-service overlaps. Continuous duct walls, formed junctions and individual terminal necks are generated from the real revised plan geometry.

Rev10 CX-08 had its lowest insulated bottom at +2.050 m, 1250 mm below this requirement. The Rev10 stacked arrangement is superseded by this roof option; it must not be installed at a 3.30 m ceiling.

DEEPEST DUCT ENVELOPE - NTS

SLAB / ROOF UNDERSIDE +4.000 m



CEILING / CLEAR HEIGHT +3.300 m

Stack: 150 mm slab / support allowance + 50 mm insulation + 400 mm bare duct + 50 mm insulation + 50 mm ceiling gap = 700 mm.

This checks scheduled duct envelopes only. Beam downstands, other services, flanges / stiffeners, damper actuators, heater casing, supports, access hatches and terminal plenums are not manufacturer-selected or surveyed.

The 50 mm gap below insulation is not a general maintenance clearance or a selected plenum dimension. Terminal drops / plenums must use the available connection space within the duct envelope; selected assemblies may require a local redesign.

User selected roof distribution / room drops. Do not cut the slab from these drawings. New penetrations, curbs, weatherproofing, roof loads and fire boundaries require structural / architectural / fire coordination.

PROPOSED LEVELS, NOT INSTALLATION APPROVAL. AFFL datum requires site confirmation. 50 mm insulation is an indoor assumption, not a verified UAE standard. D-022 identifies all proposed roof drops; actual roof routing and equipment envelopes remain open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-020

SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 20 OF 22

ADMINISTRATION BUILDING - REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

PROPOSED TERMINAL MOVES FROM REV10

Terminal	Rev10 X, Y mm	Rev11 X, Y mm	L/s
SD-02-01	2700, 18000	2700, 17700	166.2
SD-02-02	7100, 18000	7100, 17700	166.2
RG-02-01	5400, 17000	2000, 17000	151.36584740090174
RG-02-03	5400, 11900	2000, 11900	151.36584740090174
SD-08-01	6500, 6500	6800, 6500	147.43333333333334
SD-08-03	6500, 4200	6800, 4200	147.43333333333334
SD-08-05	6500, 1700	6800, 1700	147.43333333333334
RG-08-01	6400, 7150	5500, 7150	137.54389826726785
RG-08-02	11500, 7150	11400, 7150	137.54389826726785
RG-08-03	5900, 4600	5500, 4600	137.54389826726785
RG-08-04	11500, 4600	11400, 4600	137.54389826726785
RG-08-05	5900, 900	5500, 900	137.54389826726785
RG-08-06	11500, 900	11400, 900	137.54389826726785

Individual duties, neck diameters and face sizes are retained. Moves make room for perimeter return trees and insulation; coordinate reflected ceiling, throw / noise, lighting and maintenance access before adoption.

TRACEABLE ROUTING / AIRFLOW

116 indoor section paths; 15 proposed roof drops. Local branches, headers and necks remain connected to their allocated room riser. Roof-main sizes / flows are separately listed as a functional basis; actual roof lengths and levels are TBC.

All indoor section velocities, plan centerline lengths and straight-duct friction are recomputed. Wider shallow sections replace the deep returns. Selected elbow / riser / transition losses and revised total fan ESP remain pending; straight friction is not a system pressure calculation.

Rev11 is a roof-distribution proposal developed for the new 3.30 m clear-height requirement. Rev10 remains preserved as superseded for this ceiling constraint. These are engineering coordination proposals; final ceiling / OEM / pressure / structural approvals are pending.

CLEAR SIZE REVISIONS FROM REV10

Section	Rev10 mm	Rev11 mm
RA-B02	600x400	650x400
RA-H02-U01	600x400	450x350
RA-H02-D01	600x400	450x350
RA-B08	500x650	850x400
RA-H08-U01	500x650	500x350
RA-H08-D01	500x650	500x350
RA-H08-D02	500x650	500x350
RA-B05	500x525	700x400
RA-H05-U01	500x525	700x400
RA-H05-U02	500x525	700x400
RA-H05-U03	500x525	700x400
RA-H05-D01	500x525	700x400
RA-B06	350x550	500x400
RA-H06-D01	350x550	500x400
RA-H05-L01	500x525	700x400
RA-H05-N01	500x525	700x400
RA-H05-E01	500x525	700x400
RA-H06-L01	350x550	500x400
RA-H06-L02	350x550	500x400
RA-H06-E01	350x550	500x400

Largest indoor bare depth: 400 mm. Outdoor common mains retain their flow / clear-size basis; weather insulation, actual roof routes and fitting / level selection are pending.

85 instrument requirements and all existing I/O duties are retained. Common heater / probes / conditional humidifier move to the proposed roof plant, with weather-protected equipment and mounting still to be selected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-021

SCALE: NTS AT A1

REVISION

11

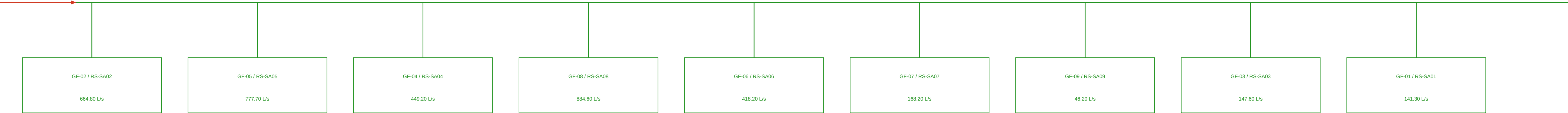
07 OCT 2026

SHEET 21 OF 22

ADMINISTRATION BUILDING - ROOF DISTRIBUTION AND ROOM RISER REGISTER

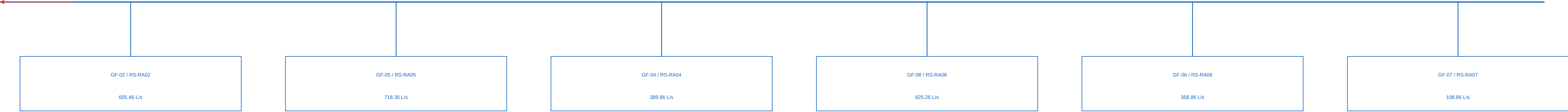
FUNCTIONAL ROOF CONNECTIONS - NTS / ACTUAL ROOF ROUTING TBC

SUPPLY FROM ROOF COMMON PLANT D-008 / 3697.80 L/s



Separate supply drops connect to each room VCD / conditional MFD / heater station; no ceiling common main.

RETURN TO ROOF COMMON PLANT D-008 / 3006.68 L/s ASSUMED



PROPOSED ROOF DROP CENTERS - SOURCE PLAN X / Y, mm

Riser	Branch / GF	X, Y mm	Clear W x H	L/s	Indoor bare BOD	Roof level / vertical length
RS-SA02	SA-B02 / 02	3600, 16600	500x350	664.80	+3.400 m	TBC / TBC
RS-RA02	RA-B02 / 02	6000, 18150	650x400	605.46	+3.400 m	TBC / TBC
RS-SA04	SA-B04 / 04	4200, 6500	350x400	449.20	+3.400 m	TBC / TBC
RS-RA04	RA-B04 / 04	4800, 8250	500x350	389.86	+3.400 m	TBC / TBC
RS-SA08	SA-B08 / 08	9800, 5600	600x400	884.60	+3.400 m	TBC / TBC
RS-RA08	RA-B08 / 08	9400, 8450	850x400	825.26	+3.400 m	TBC / TBC
RS-SA05	SA-B05 / 05	12500, 12800	600x350	777.70	+3.400 m	TBC / TBC
RS-RA05	RA-B05 / 05	19500, 10400	700x400	718.36	+3.400 m	TBC / TBC
RS-SA06	SA-B06 / 06	12500, 9475	450x300	418.20	+3.400 m	TBC / TBC
RS-RA06	RA-B06 / 06	11300, 9800	500x400	358.86	+3.400 m	TBC / TBC
RS-SA09	SA-B09 / 09	13400, 8200	200x150	46.20	+3.400 m	TBC / TBC
RS-SA07	SA-B07 / 07	15500, 5000	300x250	168.20	+3.400 m	TBC / TBC
RS-RA07	RA-B07 / 07	13000, 5000	350x250	108.86	+3.400 m	TBC / TBC
RS-SA03	SA-B03 / 03	17000, 7000	300x200	147.60	+3.400 m	TBC / TBC
RS-SA01	SA-B01 / 01	19400, 3500	300x200	141.30	+3.400 m	TBC / TBC

Roof flow/size basis is functional, not a fabricated plan: no roof survey, beam map, selected curb / sleeve / fire assembly or common heater weather enclosure was supplied. All 15 penetrations are proposals; coordinate spacing, framing, access and roof drainage before setting them out.

USER SELECTED ROOF DISTRIBUTION / ROOM DROPS. Do not treat this functional schematic as a physical roof routing / clash clearance drawing. The indoor plan is geometrically checked; roof branches, levels, insulation, lengths, losses, support loads and OEM clearances remain to be designed from site information.

TAQA TRANSMISSION N-19082-RO-012 PUMPING STATION STANDARDIZATION ADMINISTRATION HVAC DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE	DRAWING TITLE ROOF DISTRIBUTION AND ROOM RISER REGISTER FOR COORDINATION CHECKED / APPROVED: PENDING	DRAWING NUMBER ADM-HVAC-D-022 SCALE: NTS AT A1	REVISION 11 07 OCT 2026 SHEET 22 OF 22
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